

Curriculum Vitae

Ziye Lin

Master of Mathematics, Biostatistics
Department of Statistics and Actuarial Science
University of Waterloo
Waterloo, ON, Canada

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August 11, 2025

Ziye Lin

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EDUCATION

University of Waterloo

Master of Mathematics, Biostatistics

Waterloo, ON, Canada

Aug 2025

- Supervisor: Changbao Wu
- GPA: 84/100
- Research area: imputation methods in causal inference
- Relevant coursework: causal inference, probability theory, survival analysis, machine learning
- Award: 2024 W.F. Forbes Entrance Award

University of Waterloo

Bachelor of Mathematics, Statistics

Waterloo, ON, Canada

Aug 2023

- GPA: 89/100
- Award: 2020 President's Scholarship of Distinction

TEACHING DOSSIER

Teaching dossier available [online](#).

TEACHING EXPERIENCE

Tutorial Leader – [STAT 230 \(Probability\)](#)

Department of Statistics and Actuarial Science, University of Waterloo

Waterloo, ON, Canada

May 2025 – Aug 2025

- Delivered tutorial sessions regularly to approximately 40-50 students
- Clarified concepts and review practice problems
- Interacted with students to answer questions and provide support

Guest Lecturer – [STAT 231 \(Statistics\)](#)

Department of Statistics and Actuarial Science, University of Waterloo

Waterloo, ON, Canada

Mar 21, 2025

- Delivered a guest lecture to a class of 28 students
- Covered Gaussian response models and model diagnostics
- View the [lecture notes](#)
- Recorded lecture available upon request

Grader – Statistics Courses

Department of Statistics and Actuarial Science, University of Waterloo

Waterloo, ON, Canada

Sep 2024 – Aug 2025

- Provided written feedback to support students' learning
- Graded assignments and exams:
 - [STAT 230 \(Probability\)](#) (Sep 2024 – Dec 2024 & May 2025 – Aug 2025)
 - [STAT 330 \(Mathematical Statistics\)](#) (Sep 2024 – Apr 2025)
 - [STAT 331 \(Applied Linear Models\)](#) (May 2025 – Aug 2025)
 - [STAT 332 \(Sampling and Experimental Design\)](#) (Jan 2025 – Apr 2025)

Math Tutor – [MATH 237 \(Calculus III for Honours Math.\)](#)

Mathematics Tutoring Centre, University of Waterloo

Waterloo, ON, Canada

Jan 2023 – Apr 2023

- Conducted one-on-one in-person tutoring sessions with students
- Clarified course material and provided strategic hints for assignments

TEACHING AND PROFESSIONAL DEVELOPMENT

Teaching Workshops

Centre for Teaching Excellence, University of Waterloo

Waterloo, ON, Canada

Jan 2025 – Jun 2025

- Participated in four professional development workshops for post-secondary teaching
- [CTE 1106](#): Explored effective feedback strategies for teaching, focusing on phrasing, timing, utility, and emotional impact. Developed practical approaches to enhance student learning and engagement through constructive feedback
- [CTE 1111](#): Focused on designing effective tutorials in STEM disciplines. Covered strategies for planning, running, and assessing tutorials with an emphasis on active learning and student engagement. Included hands-on development of tutorial plans and peer feedback
- [CTE 1217](#): Explored the principles and benefits of active learning. Gained practical strategies and a toolkit of activities to increase student engagement, compare active learning with traditional lecturing, and design and implement interactive classroom activities
- [CTE 1234](#): Examined common causes of low teaching confidence among graduate instructors and explored mindset shifts and practical strategies to strengthen confidence and effectiveness in the classroom

MATH 900 – University Mathematics Teaching Techniques

Faculty of Mathematics, University of Waterloo

Waterloo, ON, Canada

Jan 2025 – Apr 2025

- Completed formal training in university-level mathematics instruction
- Studied course design, lecture effectiveness, assessment strategies, and student engagement
- Delivered three peer-reviewed teaching sessions on probability theory and statistical modeling
- Delivered a guest lecture to a live undergraduate class: [STAT 231 – Statistics](#)
- Designed an assignment question with marking rubric
- Authored a formal teaching statement
- By completion, qualified to be appointed as a graduate instructor in the Department of Statistics and Actuarial Science for undergraduate courses

RESEARCH AND PROJECTS

Imputation-based Approaches to Causal Inference

Master's Research Paper, University of Waterloo

Waterloo, ON, Canada

May 2025 – Aug 2025

Supervisor: Changbao Wu

- Explored imputation approaches in causal inference
- Reviewed Propensity Score Matching, Inverse Probability Weighted Estimators, AIPWE
- Investigated the performance of regression and nearest neighbor imputation methods (in R)

Modeling the Observed Radial Velocity of NGC 7531 Galaxy

Advanced Regression Project, University of Waterloo

Waterloo, ON, Canada

Jul 2023 – Aug 2023

- Modeled the association between the observed radial velocity and the position in the sky from which it is observed for NGC 7531
- Applied ordinary least squares, regularized least squares with ℓ^2 -norm penalty, cubic polynomials, and cubic B-splines (in R)
- Assessed model performances by AIC, training error, and estimated mean prediction squared error via cross-validation
- Source Dataset: [The Elements of Statistical Learning](#) → [Data](#) → [Galaxy](#)
- [Project Report](#) and [Appendix \(Code\)](#)